

# Wanda L. McCrae

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## PROFESSIONAL SUMMARY

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Full-stack developer with strong experience in Python, SQL, and REST API design. Proficient in React, Django, and FastAPI. Background in hardware engineering provides a rigorous systems-thinking perspective. Emphasis on accessibility, security, and CI/CD best practices. Excellent communicator with experience in cross-departmental and fully remote collaboration.

## TECHNICAL SKILLS

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**Programming:** Python 3, JavaScript ES6+, SQL, HTML5, CSS, Docker, Agile Methodology

**AI Tools:** Claude Code, DeepSeek

**System Design:** Microservices, REST APIs, Domain-driven design, Monoliths, Message Passing, SaaS

**API Security:** OAuth

**Front-End:** React, React Hooks, DOM Manipulation, Bootstrap, Tailwind CSS, Chakra UI

**Back-End:** Django 4, FastAPI, PostgreSQL, RabbitMQ, Twilio, SQLAlchemy, JSON

## PROFESSIONAL EXPERIENCE

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**Backend Web Developer | Tow Zone Alerts (Startup)** | towzonealerts.com 2023 - Present

- Spearheaded the design of a scalable microservices backend architecture and account services API, improving team alignment and accelerating development velocity.
- Architected a secure FastAPI microservice with PostgreSQL to manage user authentication and account data, ensuring system reliability for a growing user base.
- Orchestrated a containerized development workflow using Docker and GitHub, standardizing environments and streamlining collaboration for a fully remote team.
- Supported React.js-based front-end development to ensure cohesive integration between backend services and the user interface.

**Software Engineering Immersive Resident | Galvanize** | New York, NY (Remote) 2023

- Provided one-on-one and small group technical and project management support to students.
- Conducted code reviews on student project work.
- Shared observations on student progress with Cohort Leads and instructors.

**UX Designer | Creative Luv** | New York, NY (Remote) 2022

- Reviewed customer website across Firefox, Chrome, and Safari on desktop and mobile to assess responsiveness across environments.
- Researched and discussed options with the customer for migrating their site to a modern WYSIWYG editor to support self-managed updates.

## INDEPENDENT PROJECTS

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**Full-Stack Developer | Scam Job Reporter** | scam-job-reporter.netlify.app 2026

- Built a full-stack web application that lets job seekers anonymously report suspected scam postings; users can submit, search, and browse reports through a React frontend connected to a Python (FastAPI) backend and PostgreSQL database, all packaged in Docker containers.
- Secured the platform end-to-end: bot submissions blocked with Google reCAPTCHA, admin tokens expire after 8 hours, login attempts are rate-limited, and no personally identifying information is stored about reporters.
- Automated testing and deployment through a GitHub Actions pipeline that runs 22+ tests across frontend and backend on every push, then deploys simultaneously to Netlify and Railway only if all tests pass.
- Designed for accessibility: dyslexia-friendly font mode, dark mode, full screen-reader compatibility, and mobile-responsive layout; user preferences persist across sessions.

**Full-Stack Developer | Headline Editor** | [wlmccrae.github.io/headline-editor-ghpage](https://wlmccrae.github.io/headline-editor-ghpage) 2024 - Present

- Deployed a full-stack app to GitHub Pages and Railway giving users access to 170+ years of NY Times archives to search by date, browse article details, and rewrite headlines live in the browser; API credentials are kept secure by routing all third-party requests through the backend.
- Built a secure Python backend API with input validation, protection against malicious URLs, and resilient error handling; 93 automated tests covering edge cases and security scenarios, deployed via a GitHub Actions pipeline that blocks releases if tests fail.
- Built a responsive, accessible React SPA with Chakra UI; includes screen reader support and a dyslexia-friendly mode that switches the font and adjusts spacing site-wide, toggled from the UI and remembered across sessions.

**Front-End Developer | Simple Weather** | [itstheweather.netlify.app](https://itstheweather.netlify.app) 2023

- Built a React SPA that coordinates three external APIs (OpenWeatherMap geocoding and weather, TimeAPI) using async/await to return real-time weather and local time for any US city or current device location.
- Implemented an accessibility-focused dyslexia mode with OpenDyslexic font, high-contrast styling, and persistent localStorage preference; supplemented with semantic HTML and ARIA attributes for screen reader compatibility.
- Achieved 50-test coverage using Jest and React Testing Library, including mocked geolocation, fetch responses, and error states; integrated a gated GitLab CI/CD pipeline that blocks deployment if any tests fail.

**Full-Stack Developer | ShapeShifters** | [wmccrae.gitlab.io/shapeshifters](https://wmccrae.gitlab.io/shapeshifters) 2023

- Designed and styled a fitness meetup SPA using React with Redux Toolkit and Tailwind CSS.
- Used FastAPI to query a PostgreSQL database and expose RESTful API routes.
- Implemented authentication allowing users to create, read, update, and delete event information.

**Front-End Developer | Lotusland Fine Art (Volunteer)** | [lotuslandfineart.com](https://lotuslandfineart.com)

- Developed and managed a multi-page fine art photography website using HTML and CSS.
- Enhanced site engagement through weekly blog post updates with meticulous coding, images, and CSS styling.
- Successfully transitioned the website to a self-hosted WordPress installation.

## EARLIER EXPERIENCE

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**Senior Hardware Engineer** | Telephonics Corporation (Farmingdale, NY)

- Designed, implemented, and shipped production firmware deployed in air traffic control and military aircraft radar systems.
- Wrote and ran unit tests to exercise production code; participated regularly in code reviews.

## EDUCATION

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**M.S. Computer Engineering** | Syracuse University

**B.S. Computer Engineering** | New Jersey Institute of Technology

**Advanced Software Engineering Certificate** | Hack Reactor